MIL-N-36305(DM)
15 October 1964

MILITARY SPECIFICATION

NEEDLES, HYPODERMIC, BLOOD COLLECTING,
DOUBLE POINTED, 20 GAGE, 12s

1. SCOPE

1.1 Scope. This specification covers double pointed, blood collecting hypodermic needles.

1.2 Classification. Needles shall be of the following sizes, as specified (see 6.1):

Size 1. - 1 inch long.

Size 2. - 1-1/2 inches long.

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:22Z
Check the source to verify that this is the current version before use.

- - ' ' .)

. APPLICABLE DOCUMENTS

... The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
GG-N-196	Needles, Hypodermic; for Luer Syringes.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up, Paperboard.

Military

MIL-P-116	Preservations, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

Fed. Std. No. 66 Steel: Chemical Composition and Hardenability.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:22Z
Check the source to verify that this is the currentversion before use.

3. REQUIREMENTS

3.1 Material.

3.1.1 Cannula. The cannula shall be fabricated from 20 gage (outside diameter 0.035 ± 0.001 inch, inside diameter 0.022 inch minimum) corrosion-resisting steel tubing, conforming to AISI 304 of Federal Standard No. 66. Tubing shall be seamless drawn or welded and drawn.

3.1.2 Hub. The hub shall be fabricated from brass or aluminum. Brass shall be nickel or chromium plated.

3.2 Construction. The needles shall be constructed in strict accordance with figure 1. Each needle shall carry a wire of corrosion-resisting steel suitable for cleaning the bore of the needle. Cleaning wire shall be not less than 1/4 inch longer than the needle.

3.2.1 Cannula. The cannula shall be a straight tube with points at both ends. One point shall be for venipuncture and the other shall be for stopper puncture. The venipuncture end shall have a regular bevel point, shall be ground concave or flat, and may have a center ridge at the distal end to strengthen the tip. The end for the stopper puncture shall have a sharp, short, beveled point with a flat grind. The venipuncture point shall be so ground at the outer end as to enhance the keenness of the tip and cutting edges. The tips of the points shall be straight and strong, and the venipuncture point shall be of maximum keenness.

3.2.2 Hub. The hub shall be so constructed that the cannula goes through, and is firmly secured along, the central longitudinal axis of the hub. Hub shall be secured to the cannula by swaging.

3.3 Adherence. The cannula shall not separate from the hub, when tested as specified in 4.4.1.

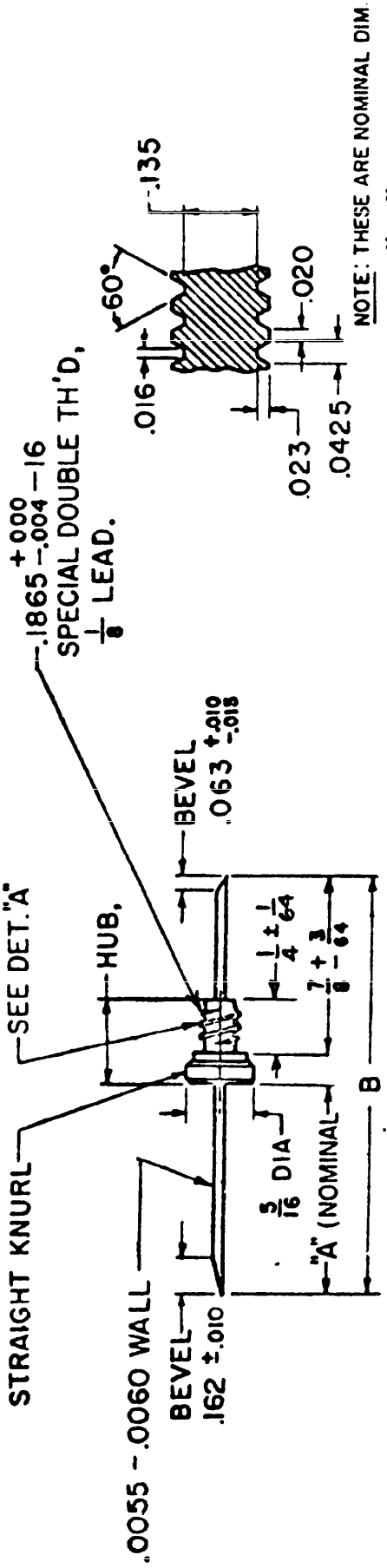
3.4 Resistance to breakage. There shall be no breakage, when the needle is tested as specified in 4.4.2.

3.5 Stiffness. The maximum deflection of the cannula shall not exceed 0.021 inch, when tested as specified in 4.4.3.

3.6 Elasticity. The cannula shall not set, when tested as specified in 4.4.4.

3.7 Corrosion resistance. The cannula shall meet the requirements for resistance to salt spray and resistance to boiling specified in GG-I-526, when tested as specified in 4.4.5 and 4.4.6.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:22Z
Check the source to verify that this is the current version before use.



DETAIL "A"
SCALE: 5/1

SIZE	A	B
1	1	$2\frac{1}{32} + \frac{1}{16}$
2	$1\frac{1}{2}$	$2\frac{17}{32} + \frac{1}{16}$

FIG. 1 NEEDLE

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:22Z
Check the source to verify that this is the current version before use.

3.8 Workmanship. Workmanship shall be first class throughout. Needles shall be clean, and shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major A defects	S-3	1.0
Major B defects	S-3	1.0
Minor A defects	S-3	2.5
Minor B defects	S-3	2.5
For dimensional examination	S-2	2.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:22Z
Check the source to verify that this is the current version before use.

[illegible])

4.2.1 [illegible] for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Unit of product shall be the needle, and inspection level [illegible].

Characteristic	Requirement	Test procedure	AQL
Adherence	3.3	4.4.1	*
Resistance to breakage	3.4	4.4.2	*
Stiffness	3.5	4.4.3	*
Elasticity	3.6	4.4.4	*
Salt-spray test	3.7	4.4.5	*
Boil test	3.7	4.4.6	*

* Acceptance number shall be zero.

4.3 Examination. The needles shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major A (7X magnification)

101 Point not sharp.
102 Point edges not free of feathering.
103 Point not free of hooks and bends.
104 Point not symmetrical.
105 Lumen not free of constriction.
106 Cannula not straight (deviation from longitudinal axis exceed 3°).
107 Cannula not free of cracks.
108 Cannula not free of scale and corrosion.
109 Cannula not free of slivers.
110 Cannula not free of foreign matter.
111 Cannula not free of burs.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:22Z
Check the source to verify that this is the currentversion before use.

Major B

151 Threading not free of breaks.
152 Hub not free of distortion.
153 Hub not free of cracks.

Minor A (7X magnification)

201 Cannula not free of pits.
202 Cannula not free of tool marks.

Minor B

231 Hub not free of fins.
232 Hub not free of foreign matter.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The needles shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Adherence. A straight pull of 15 pounds shall be applied between the cannula and the hub for a period of 1 minute.

4.4.2 Resistance to breakage. The needle hub shall be held in a fixed position, and a force shall be applied to the distal end of the cannula, 1 inch from the hub, in order to bend the cannula in a clockwise direction through an arc of 25° from the original or neutral position. The operation shall then be reversed (i.e., bent through an arc of 25° from the neutral position in a counterclockwise direction) and finally, the cannula shall be returned to the neutral position. This complete operation (or cycle) shall be repeated 20 times.

4.4.3 Stiffness. The cannula shall be placed across two test edges (0.002-inch radius maximum) 0.750 inch $\pm$ 0.003 inch apart. A load of 50 ounces $\pm$ 1/4 ounce shall be applied to the center of the span, and the resulting maximum deflection of the cannula shall be measured using a gage whose minimum graduations are not greater than 0.001 inch. The force shall be applied through an edge with a maximum radius of 0.036 inch, and shall not be farther than 0.003 inch from the center.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:22Z
Check the source to verify that this is the current version before use.

Major B

151 Threading not free of breaks.
152 Hub not free of distortion.
153 Hub not free of cracks.

Minor A (7X magnification)

201 Cannula not free of pits.
202 Cannula not free of tool marks.

Minor B

231 Hub not free of fins.
232 Hub not free of foreign matter.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The needles shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Adherence. A straight pull of 15 pounds shall be applied between the cannula and the hub for a period of 1 minute.

4.4.2 Resistance to breakage. The needle hub shall be held in a fixed position, and a force shall be applied to the distal end of the cannula, 1 inch from the hub, in order to bend the cannula in a clockwise direction through an arc of 25° from the original or neutral position. The operation shall then be reversed (i.e., bent through an arc of 25° from the neutral position in a counterclockwise direction) and finally, the cannula shall be returned to the neutral position. This complete operation (or cycle) shall be repeated 20 times.

4.4.3 Stiffness. The cannula shall be placed across two test edges (0.002-inch radius maximum) 0.750 inch $\pm$ 0.003 inch apart. A load of 50 ounces $\pm$ 1/4 ounce shall be applied to the center of the span, and the resulting maximum deflection of the cannula shall be measured using a gage whose minimum graduations are not greater than 0.001 inch. The force shall be applied through an edge with a maximum radius of 0.036 inch, and shall not be farther than 0.003 inch from the center.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:22Z
Check the source to verify that this is the currentversion before use.

4.4.4 Elasticity. Elasticity shall be tested in accordance with [illegible]-196.

4.4.5 Salt-spray test. Salt-spray test shall be performed as specified in GG-I-526.

4.4.6 Boil test. Boil test shall be performed as specified in GG-I-526.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Twelve (12) needles, of the size specified, constitute one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	576 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Twelve (12) needles shall be assembled on a substantial cardboard, plastic, or metal rack, so designed that each individual needle is held securely in place in such a manner that the points cannot be damaged by contact with any part of other needles or container. Each rack (1 unit) shall be further packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commerical colors shall be acceptable. Closure shall be adequate to prevent spillage of contents under normal handling. Each carton shall be further processed in accordance with method IA-8 of MIL-P-116.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:22Z
Check the source to verify that this is the current version before use.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PP-B-676, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 Level B.

5.1.4.1 Unit package. Twelve (12) needles shall be assembled on a substantial cardboard, plastic, or metal rack, so designed that each individual needle is held securely in place in such a manner that the points cannot be damaged by contact with any part of other needles or container. Each rack (1 unit) shall be further packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spillage of contents under normal handling.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of PPP-B-636. In addition, waterproofing shall be in accordance with 30.4.2 of appendix to box specification.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receving activity within the continental limits of the United States for storage and redistribution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:22Z
Check the source to verify that this is the current version before use.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure of fiberboard box shall be in accordance with method I of appendix to box specification.

5.3 Marking.

5.3.1 Interior packages. Each unit carton and barrier (when used) and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown. Waterproofing of labels shall be required only when carton utilized is fabricated of water-resistant material.

5.3.2 Exterior containers. Exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:22Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

FORM APPROVED BUDGET BUREAU NO. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: MILITARY SPECIFICATION MIL-N-36305(DM), dated 15 October 1964
NEEDLES, HYPODERMIC, BLOOD COLLECTING, DOUBLE POINTED, 20 GAGE, 12s

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (ATTACH ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS SPECIFICATION. IF THERE ARE ADDITIONAL PAPERS, ATTACH TO FORM AND PLACE BOTH IN AN ENVELOPE ADDRESSED TO PREPARING ACTIVITY)

SUBMITTED BY (PRINTED OR TYPED NAME AND ACTIVITY)	DATE

DD FORM 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:22Z
Check the source to verify that this is the current version before use.